

Jason Bowman

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SUMMARY

Staff/Principal-level engineer, by turns hacker, maker, and academic. Over a decade scaling production systems from startup to IPO at Uber, Unity, and Highnote. Years of carrying the pager forged one conviction: correctness, resilience, and integrity are designed in, not bolted on. Infrastructure must be **correct by construction, not by prayer and runbook**, grounded in computer-science fundamentals, not guesswork. That conviction is now full-time and in the open: co-founding *Denful*, an open-source organization building composable, verifiable foundations that make rigor the default, from a single laptop to a fleet of thousands. That rigor only matters more as agentic coding makes plausible-but-wrong systems cheap to produce and costly to verify.

EDUCATION

The University of Alabama

Master of Science in Computer Science (MSc)

Bachelor of Science in Computer Science (BSc), Minor in Mathematics

Tuscaloosa, AL

2012

2011

SKILLS

Languages & Platforms: Scala, Nix, Python, Go, Java, C/C++, Ruby, and UNIX shell; 25+ years on GNU/Linux and UNIX; strong functional-programming background

Language Engineering & Tooling: DSL design, type systems, algebraic effects, term and graph rewriting, AST-based codemods and automated migration, parser generators (ANTLR, tree-sitter), macros and metaprogramming; sbt, Bloop, Metals, JMH, Gatling

Distributed Systems: Kafka, stream processing, high-QPS services in hyper-growth environments, elastic compute, API design, performance optimization, graceful fault detection and recovery

Cloud & Data: GCP and AWS – BigQuery, BigTable, Spanner, Dataflow / Apache Beam / Scio for specialized real-time pipelines, Avro (avro4s), GKE; PostgreSQL, Redis, ZooKeeper, Hadoop, Vertica

Systems: Linux kernel and kernel-module development, device drivers, network-stack and protocol optimization, source-level performance tuning, multimedia (gstreamer, ffmpeg)

AI & Machine Learning: Reinforcement learning (PPO, SAC, GAIL) and distributed RL training platforms (pre-LLM-boom); LLM agentic workflows, multi-agent orchestration, and spec-driven development

Kubernetes & Cloud-Native: Cilium (eBPF CNI and network policies), microVMs, hardware virtualization (VFIO, GPU passthrough)

GitOps & Infrastructure-as-Code: ArgoCD, Flux, Terraform, Ansible, Puppet, Chef, SaltStack, bare-metal and cloud fleet management, declarative secrets (agenix, SOPS), CI/CD and developer-experience tooling

Observability & Reliability: Grafana, Prometheus, Grafana Alloy, metrics and log pipelines, alerting; 24/7 on-call, automated incident detection and remediation, backup and disaster recovery

Security & Compliance: security auditing, applied cryptography (KMS envelope encryption, AES-CBC, key derivation), OIDC and access control, PII/GDPR data governance, HIPAA compliance

Leadership: Technical Project Management, Team Leadership, Mentorship, Organizational Leadership

EXPERIENCE

Denful

Co-Founder

Open Source

2023 - Present

- Co-founded an open-source organization building composable, aspect-oriented Nix tooling that makes system configuration verifiable and correct by construction, grounded in algebraic effects, interaction nets, and graph rewriting, with an active and growing community (*den* at 450+ GitHub stars).
- **gen**: Authored *gen*, the ecosystem's foundational layer: a suite of dependency-light, pure-Nix libraries spanning algebraic primitives, higher-order scope graphs, typed registries, selector and graph-query algebras, and guarded graph rewriting, the small, independently testable substrate the rest of the ecosystem rests on.
- **den**: Lead developer of *den*, a context-aware, aspect-oriented configuration framework that produces NixOS, Darwin, and Home-Manager outputs from a single declarative source, using algebraic effects to make resolution explicit, testable, and traceable.
- **Domain-Agnostic Configuration**: Generalized the engine beyond NixOS to arbitrary configuration targets, demonstrated by extending it to Kubernetes GitOps through upstream contributions to *nixidy*: a generic object match/rewrite rule engine and CRD-native module accessors.
- **Graph-Based Infrastructure**: Modeled service topology, firewall rules, and access controls as derived properties of a typed relationship graph, synthesizing network policies and ACLs from declared service relationships rather than hand-maintaining configuration.

Highnote

Senior Software Engineer

San Francisco, CA

2022 - 2023

- Infrastructure engineering for a B2B card-issuing fintech, focused on data systems and large-scale platform migrations.
- **Spanner Replication System**: Implemented a transaction-log replay and mirroring system on Google Dataflow for cross-region Spanner replication, sustaining 99.99% uptime for payment processing.
- **Codemods & Migration Tooling**: Engineered AST-based codemods and automated migration utilities driving mono-repo-wide framework upgrades across 50+ microservices, cutting manual effort ~80% and eliminating version skew.
- Delivered data pipeline infrastructure for real-time payment analytics and fraud detection, processing millions of transactions daily with sub-100ms latency requirements.

Unity Technologies

Senior Software Engineer / Tech Lead

San Francisco, CA

2018 - 2022

- Led engineering teams and architected large-scale data platform infrastructure serving analytics, advertising, and ML workloads at one of the world's largest mobile ad networks.
- **Data Platform (Sole Architect)**: Architect and primary engineer of Unity's company-wide event-ingestion platform – the production replacement for a legacy daily-batch flow, built during the AWS-to-GCP migration while leading a team of 6. On Apache Beam/Scio and Dataflow (Kafka → schema'd Avro → GCS → BigQuery), it sustained 1M events/sec (10B/day) and cut end-to-end data-availability latency from 24+ hours to under 15 minutes at 99.99% availability, consolidating three previously independent pipelines (analytics, advertising, CDP) into a single platform adopted across the company.
- **GDPR & CCPA Compliance**: Architected the platform's privacy layer and led the cross-functional rollout with Legal and company-wide data owners. PII was split into a separate KMS-encrypted stream and access-controlled store, with subject-level cryptographic compliance keys enabling right-to-erasure across the data lake – plus PII hashing, geo-coordinate fuzzing, GeolIP enrichment,

and automated statistical PII classification – flagging fields by value cardinality and uniqueness for data-owner review and refinement. Per-owner bucketing provided tenant-scoped access controls and cost chargeback.

- **Runtime Control Plane:** Designed a live control plane – BigTable-backed schema, routing, and encoding registries hot-reloaded into running pipelines – so data owners onboarded new event types and evolved schemas with zero redeloys, decoupling platform operations from per-team data changes. Automated schema derivation inferred Avro contracts from existing data.
- **Reliability Tier:** Built the platform's fault-tolerance layer – dead-letter capture/replay with rate-limited backoff and BigQuery load-job monitoring with automatic retry – sustaining delivery guarantees at 10B events/day.
- **ML-Agents RL Platform:** Architected a cloud-native reinforcement-learning platform on GKE for Unity's ML-Agents – applied RL (PPO, SAC, GAIL) years before the current AI boom – orchestrating distributed training experiments and Unity simulation builds, and scaling researchers' workloads across hundreds of nodes.
- **Unity Simulation & Robotics:** Stood up remote-tunneling infrastructure enabling secure access to Unity instances in shared cloud environments. Developed tooling to bridge on-premises robotics workflows with cloud-based simulation.

Uber Technologies, Inc

San Francisco, CA

Software Engineer

2015 - 2018

Marketplace Engineering // Fares

2017 - 2018

- Led development of end-user-facing product features spanning back-end services and mobile applications.
- **Pricing Integrity Engine:** Led development of a real-time and batch data-validation platform for fare data, with configurable business rules, analytics, and automatic remediation.
- **Fares Platform:** Designed next-generation fare calculation platform for all Uber products using a descriptive DSL, integrating tolls, maps, surge, and subscription data.
- **Promotions Platform:** Created auditing and remediation tools for post-transaction analysis of user promotions across real-time systems.

Site Reliability Engineering

2015 - 2017

- An early engineer at Uber and founding engineer of its Site Reliability Engineering organization. Originally served all of Data – the better part of a year on production-critical Vertica and data-warehouse reliability (30% of Uber's hardware footprint), automating provisioning, replication, ETL, and cluster management across Vertica, Hadoop, Redis, and MemSQL. Then, as a lead on the Streaming SRE team, scaled one of the world's largest Kafka deployments of its era to thousands of brokers at >500 GiB/s across multiple regions.
- **Kafka Infrastructure:** Built an intelligent partition-redistribution engine spanning hundreds of brokers across many clusters, driving load-distribution standard deviation from ~53% to 0.0001% of mean, eliminating disk hotspots and self-healing broker failures. Kernel, filesystem, JVM, and broker-source-code tuning delivered over 5× (500%) throughput in synthetic benchmarks; in production, p99 produce latency dropped from recurring ~500ms spikes to a stable ~10ms (~50×) and request-handler utilization fell from ~80% to ~20% – averting a projected nine-figure migration to an all-SSD broker fleet by proving the existing disk stack could meet I/O targets through optimization alone.
- **uServer:** Single-handedly built a self-service bare-metal provisioning platform adopted company-wide, managing hundreds of thousands of hosts across Uber's datacenters.
- Designed data security architecture and compliance tooling for global expansion into China, India, and Russia.

Rocket Lawyer

San Francisco, CA

Senior Systems Engineer (original title: Systems Engineer)

2013 - 2015

- Designed service infrastructure and provided 24/7 on-call support for a high-volume legal-services e-commerce site running primarily on the JVM and a legacy .NET platform. Improved reliability from 99% uptime to 99.99% and cut p99 latency from 2000ms to 150ms.
- Analyzed and optimized the performance of application services and the supporting infrastructure tier.
- Instrumented application code to expose internal state and key performance metrics, enrich logging, and reduce overall latency. Implemented dynamic service discovery and reconfiguration via ZooKeeper.
- Built tooling for continuous integration, dynamic configuration management, deployment management, monitoring, and automated incident detection and remediation.

nine.is

Software Engineer / Web Developer

Tuscaloosa, AL
2012

- Full-stack development and team leadership for a design company expanding into in-house software. Built applications using PHP, JavaScript, and Ruby on Rails.

The University of Alabama

Graduate Research Assistant

Tuscaloosa, AL
2010 - 2012

- Export-controlled research on cross-layer network optimization and security for next-generation internet infrastructure. Worked in the Linux networking stack – custom TCP-variant routing protocols and kernel-module development for fine-grained tuning and performance analysis – alongside an adaptive video-streaming client (gstreamer), automated VM-image penetration testing, and security analysis for the GENI experimental network testbed.

Brewer Porch Children's Center

Unix Systems Administrator

Tuscaloosa, AL
2004 - 2006

- Systems administration for a healthcare organization. Managed Linux/Solaris infrastructure and ensured HIPAA compliance for patient data.

PUBLICATIONS

- Dawei Li, Jason Bowman, Xiaoyan Hong "Evaluation of Security Vulnerabilities by Using ProtoGENI as a Launchpad", IEEE Globecom 2011, Houston, USA, Dec. 2011.
- M. Anderson, P. Kilgo, and J. Bowman. "RDIS: Generalizing domain concepts to specify device to framework mappings." In International Conference on Robotics and Automation, 2012.
<http://ua-robotics.net/index.php?title=RDIS>

OUTREACH AND COMMUNITY INVOLVEMENT

- Mentor for Hackbright Academy, a 12 week intensive coding bootcamp focused on empowering women aiming to pursue a career in the software engineering field.
- Mentor for dev/Mission, a non-profit program that exposes low-income youth aged 16-24 to careers in technology.
- Organized and ran an introductory Scala and Functional Programming course based on Martin Odersky's course and book. Resources available on my personal github.
- Served as president from 2010 - 2012 of the Association for Computing Machinery student chapter at The University of Alabama.
- Organized a mentoring/tutoring program for freshman and sophomore computer science students to improve program retention and graduation rates. Still going to this day, and graduation rates of freshman classes are up from 5% to nearly 50%.

INTERESTS

- **Professional:** Functional Programming · Language Development · Generative Programming · Operating Systems · System Architecture · Distributed Systems · Stream Processing · Virtualization · Open Source · GNU/Linux
- **Personal:** Reverse Engineering · Embedded Systems · Electrical Engineering · Robotics · Game Development & Theory · Music Production · Fabrication and DIY · CAD Design · Privacy · Mechanical Keyboards · Rock Climbing · Emacs